

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT – 3 – FALL SEMESTER 2026-2027

Programme Name & Branch: M.Tech. (Integrated) CSE (Data Science)

Course Name: Data Science Programming Lab

Course Code: CSI3004

Assessment 3:

Customer Segmentation

Problem Statement

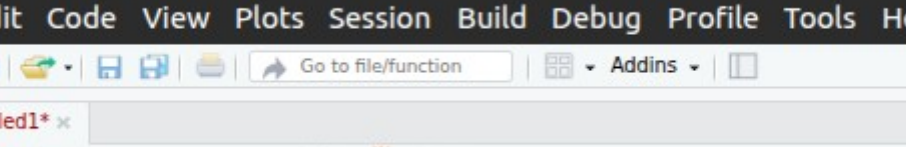
A company wants to segment its customers based on demographic, psychographic, and behavioral patterns using R.

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1. Data Understanding:

- Load the dataset
- Display structure and summary
- Identify variables and their types



The screenshot displays the RStudio application window. At the top is a dark menu bar with the following items: File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu bar is a toolbar containing icons for file operations (new, open, save, save all, print), a search bar labeled 'Go to file/function', and other utility icons. The main workspace area shows a script editor with a tab labeled 'Untitled1*'. The script contains the following R code:

```
1 install.packages("ggplot2")
2 library(ggplot2)
3 customers <- read.csv("~/Downloads/Mall_Customers.csv")
4 head(customers)
5 str(customers)
6 summary(customers)
7 names(customers)
8 sapply(customers, class)
9
```

2)Data Preparation :

- Scale the dataset

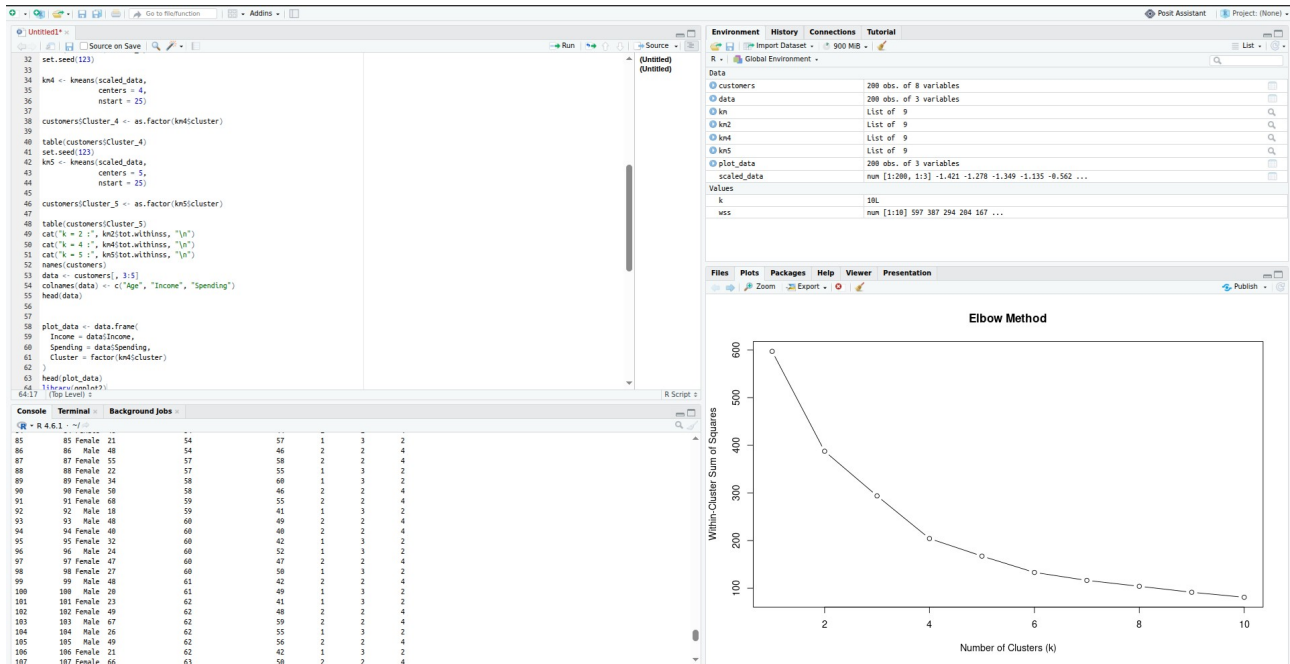
```

9
10 data <- customers[, 3:5]
11 head(data)
12 scaled_data <- scale(data)
13 head(scaled_data)
14

```

3. Clustering :

- Use Elbow Method to find optimal clusters



- Apply K-Means clustering
- Assign cluster labels

```

16 wss <- c()
17 for (k in 1:10) {
18   km <- kmeans(scaled_data, centers = k, nstart = 25)
19   wss[k] <- km$tot.withinss
20 }
21 plot(1:10, wss,
22      type = "b",
23      xlab = "Number of Clusters (k)",
24      ylab = "Within-Cluster Sum of Squares",
25      main = "Elbow Method")

```

4. Visualization :

- Plot clusters using ggplot2:

```
plot_data <- data.frame(
  Income = data$Income,
  Spending = data$Spending,
  Cluster = factor(km4$cluster)
)
head(plot_data)
library(ggplot2)
ggplot(plot_data,
  aes(x = Income,
      y = Spending,
      color = Cluster)) +
  geom_point(size = 3) +
  labs(
    title = "Customer Segmentation using K-Means",
    x = "Annual Income (k$)",
    y = "Spending Score (1-100)",
    color = "Cluster"
  ) +
  theme_minimal()
```

- Interpret cluster differences :

INTERPRET DIFFERENCES:

Cluster difference. The K-Means algorithm groups customers based on similarities in age, annual income, and spending score. The clusters differ in their income and spending behaviour. Some clusters represent customers with lower income and lower spending, while others represent customers with higher income and higher spending. There are also customers with high income but relatively low spending and customers with lower income but relatively high spending. These differences help the company identify different customer segments and design suitable marketing strategies for each group.

5. Experimentation :

- Try different number of clusters (k = 2, 4, 5)

```
26 set.seed(123)
27 km2 <- kmeans(scaled_data,
28               centers = 2,
29               nstart = 25)
30 customers$Cluster_2 <- as.factor(km2$cluster)
31 table(customers$Cluster_2)
32 set.seed(123)
33
34 km4 <- kmeans(scaled_data,
35               centers = 4,
36               nstart = 25)
37
38 customers$Cluster_4 <- as.factor(km4$cluster)
39
40 table(customers$Cluster_4)
41 set.seed(123)
42 km5 <- kmeans(scaled_data,
43               centers = 5,
44               nstart = 25)
45
46 customers$Cluster_5 <- as.factor(km5$cluster)
47
```

- Compare results:

```
table(customers$Cluster_5)
cat("k = 2 :", km2$tot.withinss, "\n")
cat("k = 4 :", km4$tot.withinss, "\n")
cat("k = 5 :", km5$tot.withinss, "\n")
names(customers)
data <- customers[, 3:5]
colnames(data) <- c("Age", "Income", "Spending")
head(data)
```

The screenshot shows an R console window with the following content:

```
R - R 4.6.1 - ~/
> # display first few records
> head(customers)
  CustomerID Genre Age Annual.Income..k.. Spending.Score..1.100.
1          1  Male  19              15              39
2          2  Male  21              15              81
3          3 Female  20              16               6
4          4 Female  23              16              77
5          5 Female  31              17              40
6          6 Female  22              17              76
>
> # 2. DATA UNDERSTANDING
> # -----
>
> str(customers)
'data.frame':  200 obs. of  5 variables:
 $ CustomerID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Genre           : chr  "Male" "Male" "Female" "Female" ...
 $ Age             : int  19 21 20 23 31 22 35 23 64 30 ...
 $ Annual.Income..k.. : int  15 15 16 16 17 17 18 18 19 19 ...
 $ Spending.Score..1.100.: int  39 81 6 77 40 76 6 94 3 72 ...
>
> summary(customers)
  CustomerID      Genre      Age      Annual.Income..k..  Spending.Score..1.100.
Min.   : 1.00  Length:200  Min.   :18.00  Min.   : 15.00  Min.   : 1.00
1st Qu.:50.75  N.unique: 2   1st Qu.:28.75  1st Qu.: 41.50  1st Qu.:34.75
Median :100.50  N.blank : 0   Median :36.00  Median : 61.50  Median :50.00
Mean   :100.50  Min.nchar: 4   Mean   :38.85  Mean   : 60.56  Mean   :50.20
3rd Qu.:150.25  Max.nchar: 6   3rd Qu.:49.00  3rd Qu.: 78.00  3rd Qu.:73.00
Max.   :200.00          Max.   :70.00  Max.   :137.00  Max.   :99.00
>
> names(customers)
[1] "CustomerID"      "Genre"           "Age"             "Annual.Income..k.." "Spending.Score..1.100."
>
> # 3. IDENTIFY VARIABLES AND THEIR TYPES
> # -----
>
> sapply(customers, class)
      CustomerID      Genre      Age      Annual.Income..k..  Spending.Score..1.100.
      "integer"    "character"  "integer"      "integer"      "integer"
>
> # 4. SELECT VARIABLES FOR CLUSTERING
> # -----
>
> data <- customers[, 3:5]
>
> head(data)
  Age Annual.Income..k..  Spending.Score..1.100.
1  19              15              39
2  21              15              81
3  20              16               6
4  23              16              77
5  31              17              40
6  22              17              76
>
> # 5. SCALE THE DATA
```

```
Source
Console Terminal Background Jobs
R v 4.6.1 ~ /
>
> # 5. SCALE THE DATA
> # -----
>
> scaled_data <- scale(data)
>
> head(scaled_data)
      Age Annual.Income..k.. Spending.Score..1.100.
[1,] -1.4210029      -1.734646      -0.4337131
[2,] -1.2778288      -1.734646      1.1927111
[3,] -1.3494159      -1.696572      -1.7116178
[4,] -1.1346547      -1.696572      1.0378135
[5,] -0.5619583      -1.658498      -0.3949887
[6,] -1.2062418      -1.658498      0.9990891
>
>
> # 6. ELBOW METHOD
> # -----
>
> wss <- c()
>
> for (k in 1:10) {
+   km <- kmeans(scaled_data, centers = k, nstart = 25)
+   wss[k] <- km$tot.withinss
+ }
>
> plot(1:10, wss,
+      type = "b",
+      xlab = "Number of Clusters (k)",
+      ylab = "Within-Cluster Sum of Squares",
+      main = "Elbow Method")
>
>
> # 7. K-MEANS FOR k = 2
> # -----
>
> set.seed(123)
>
> km2 <- kmeans(scaled_data,
+               centers = 2,
+               nstart = 25)
>
> customers$Cluster_2 <- as.factor(km2$cluster)
>
> table(customers$Cluster_2)
 1  2
97 103
>
>
> # 8. K-MEANS FOR k = 4
> # -----
>
> set.seed(123)
>
> km4 <- kmeans(scaled_data,
+               centers = 4,
+               nstart = 25)
>
```

```
Source
Console Terminal Background Jobs
R 4.6.1 ~/  
1 2 3 4  
38 65 57 40  
>  
>  
> # 9. K-MEANS FOR k = 5  
> # -----  
>  
> set.seed(123)  
>  
> km5 <- kmeans(scaled_data,  
+               centers = 5,  
+               nstart = 25)  
> customers$Cluster_5 <- as.factor(km5$cluster)  
>  
> table(customers$Cluster_5)  
  
1 2 3 4 5  
40 54 39 47 20  
>  
>  
> # 10. COMPARE RESULTS  
> # -----  
>  
> cat("k = 2 :", km2$tot.withinss, "\n")  
k = 2 : 387.4393  
> cat("k = 4 :", km4$tot.withinss, "\n")  
k = 4 : 204.199  
> cat("k = 5 :", km5$tot.withinss, "\n")  
k = 5 : 167.4063  
>  
>  
> # 12. CLUSTER SUMMARY  
> # -----  
>  
> names(customers)  
[1] "CustomerID"      "Genre"            "Age"              "Annual.Income..k.." "Spending.Score..1.100." "Cluster_2"  
[7] "Cluster_4"       "Cluster_5"  
>  
> data <- customers[, 3:5]  
>  
> colnames(data) <- c("Age", "Income", "Spending")  
>  
> head(data)  
Age Income Spending  
1 19 15 39  
2 21 15 81  
3 20 16 6  
4 23 16 77  
5 31 17 40  
6 22 17 76  
>  
>  
> # VISUALIZATION USING GGPLOT2  
> # -----  
>  
> plot_data <- data.frame(  
+   Income = data$Income..
```

```

Source
Console Terminal Background Jobs
R 4.6.1 ~ /
85 85 Female 21 54 57 1 3 2
86 86 Male 48 54 46 2 2 4
87 87 Female 55 57 58 2 2 4
88 88 Female 22 57 55 1 3 2
89 89 Female 34 58 60 1 3 2
90 90 Female 50 58 46 2 2 4
91 91 Female 68 59 55 2 2 4
92 92 Male 18 59 41 1 3 2
93 93 Male 48 60 49 2 2 4
94 94 Female 40 60 40 2 2 4
95 95 Female 32 60 42 1 3 2
96 96 Male 24 60 52 1 3 2
97 97 Female 47 60 47 2 2 4
98 98 Female 27 60 50 1 3 2
99 99 Male 48 61 42 2 2 4
100 100 Male 20 61 49 1 3 2
101 101 Female 23 62 41 1 3 2
102 102 Female 49 62 48 2 2 4
103 103 Male 67 62 59 2 2 4
104 104 Male 26 62 55 1 3 2
105 105 Male 49 62 56 2 2 4
106 106 Female 21 62 42 1 3 2
107 107 Female 66 63 50 2 2 4
108 108 Male 54 63 46 2 2 4
109 109 Male 68 63 43 2 2 4
110 110 Male 66 63 48 2 2 4
111 111 Male 65 63 52 2 2 4
112 112 Female 19 63 54 1 3 2
113 113 Female 38 64 42 2 1 3
114 114 Male 19 64 46 1 3 2
115 115 Female 18 65 48 1 3 2
116 116 Female 19 65 50 1 3 2
117 117 Female 63 65 43 2 2 4
118 118 Female 49 65 59 2 2 4
119 119 Female 51 67 43 2 2 4
120 120 Female 50 67 57 2 2 4
121 121 Male 27 67 56 1 3 2
122 122 Female 38 67 40 2 1 3
123 123 Female 40 69 58 1 4 1
124 124 Male 39 69 91 1 4 1
125 125 Female 23 70 29 1 1 3
[ reached 'max' / getOption("max.print") -- omitted 75 rows ]
>
>
> # INTERPRETATION OF SEGMENTS
> # -----
>
> aggregate(
+ data,
+ by = list(Cluster = customers$Cluster_4),
+ FUN = mean
+ )
Cluster Age Income Spending
1 1 39.36842 86.50000 19.57895
2 2 53.98462 47.70769 39.96923
3 3 25.43860 40.00000 60.29825
4 4 32.87500 86.10000 81.52500
>

```

6.Expected Output :

- Clustered dataset

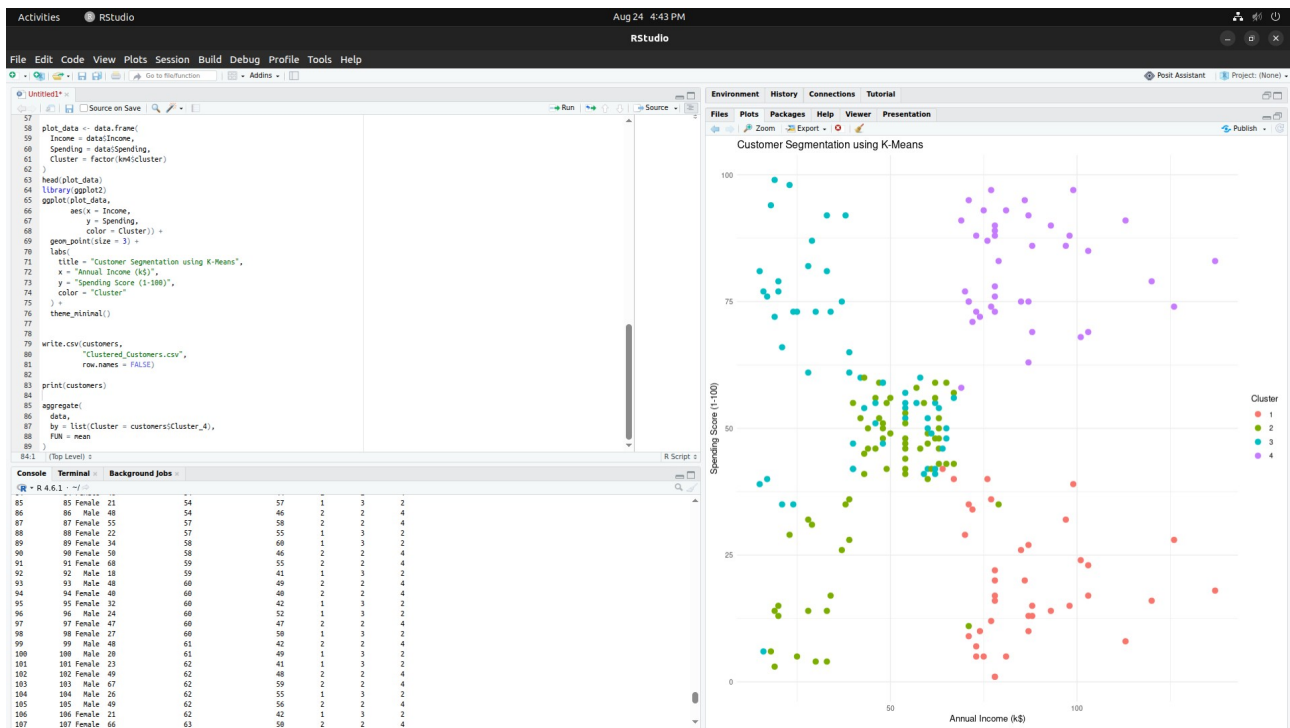
```

write.csv(customers,
          "Clustered_Customers.csv",
          row.names = FALSE)

print(customers)

```

- Graph visualization :



- Interpretation of segments:

```

84
85 aggregate(
86   data,
87   by = list(Cluster = customers$Cluster_4),
88   FUN = mean
89 )

```